4.1.1 Charge

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	electric current as rate of flow of charge; $I = \frac{\Delta Q}{\Delta t}$	
(b)	the coulomb as the unit of charge	
(c)	the elementary charge *e* equals 1.6×10^{-19} C	Learners will be expected to know that an electron has charge –*e* and a proton a charge +*e*. HSW7
(d)	net charge on a particle or an object is quantised and a multiple of *e*	
(e)	current as the movement of electrons in metals and movement of ions in electrolytes	HSW7
(f)	conventional current and electron flow	HSW7
(g)	Kirchhoff's first law; conservation of charge.	